MASTER OF COMPUTER SCIENCE
BINUS GRADUATE PROGRAM
BINA NUSANTARA UNIVERSITY
JAKARTA

ASSESSMENT FORM

Course: COMP8047041 - Business Intelligence and Analytics

Method of Assessment: Project

Semester/Academic Year : 1/2025-2026

Name of Lecturer : Dr. Devi Fitrianah, S.Kom., M.T.I

Date : ………………………

Class : ………………………

Topic : Business Intelligence dashboard

Group Members :	1____________________ 2____________________ 3____________________ 4____________________ 5____________________ 6____________________ 7____________________ 8____________________

Student Outcomes:

(SO 5) Mampu mengembangkan ilmu pengetahuan dan Teknologi Informasi dan Komunikasi dengan menggunakan metode kecerdasan buatan untuk menghasilkan produk-produk inovatif yang dapat diaplikasikan dalam berbagai bidang.
Able to develop science and Information and Communication Technology using artificial intelligence methods to produce innovative products that can be applied in various fields.

Learning Objectives:

(L.Obj 5.2) Mampu menggabungkan serta mengembangkan penelitian dalam komputasi intelegensia sesuai dengan kebutuhan industri.
Able to combine and develop research in computational intelligence in accordance with the needs of industry.

Learning Outcomes:

- LO2: Design a dimensional and physical model for data warehouse.(L.Obj 5.2)
- LO3: Comprehend ETL strategies (L.Obj 5.2)
- LO4: Design and develop business intelligence applications.(L.Obj 5.2)

No	Related LOLOBJ-SO	Assessment criteria	Weight	Excellent (90 - 100)	Very Good (85-89)	Good (80-84)	Average (75-79)	Poor (0-74)	Score	(Score x Weight)
1	LO2	Kemampuan mendesain model dimensional dan fisik untuk data warehouse *Ability to design a dimensional and physical model for a data warehouse*	30%	Desain dimensional & fisik sangat tepat, lengkap, dan optimal, mencakup semua fakta & dimensi relevan, hubungan antar entitas benar, spesifikasi fisik (tipe data, indeks, partisi) detail, serta mempertimbangkan performa dan integritas data secara menyeluruh *The dimensional & physical design*	Desain dimensional & fisik tepat dan lengkap dengan sedikit kekurangan minor pada detail hubungan, spesifikasi fisik, atau dokumentasi. *The dimensional & physical design is accurate and complete with only minor shortcomings in relationship details, physical specifications,*	Desain mencakup sebagian besar fakta & dimensi, hubungan antar entitas umumnya benar, namun ada beberapa kekurangan pada spesifikasi fisik atau integritas data. *The design covers most facts & dimensions, relationships are generally correct, but some deficiencies exist in physical specifications or*	Desain kurang lengkap, beberapa fakta/dimensi penting terlewat, hubungan antar entitas atau spesifikasi fisik kurang tepat. *The design is incomplete, missing several important facts/dimensions, and has inaccuracies in entity relationships or physical specifications.*	Desain tidak sesuai konsep dimensional & fisik data warehouse, banyak elemen penting hilang atau salah, tidak dapat digunakan. *The design does not follow the dimensional & physical data warehouse concept, with many critical elements missing or incorrect,*		

No	Related LOLOBJ-SO	Assessment criteria	Weight	Excellent (90 - 100)	Very Good (85-89)	Good (80-84)	Average (75-79)	Poor (0-74)	Score	(Score x Weight)
				is highly accurate, complete, and optimal, covering all relevant facts & dimensions, with correct entity relationships, detailed physical specifications (data types, indexes, partitions), and full consideration of performance and data integrity.	*or documentation.*	*data integrity considerations.*		*rendering it unusable.*		
2	LO3	Kemampuan memahami strategi ETL (Extract, Transform, Load)	30%	Menunjukkan pemahaman yang sangat mendalam dan komprehensif tentang konsep, proses, dan strategi ETL, termasuk teknik lanjutan, metode optimisasi, penanganan kualitas data, serta pemilihan tools, dan mampu menjelaskan	Menunjukkan pemahaman yang jelas dan akurat tentang konsep, proses, dan strategi ETL, dengan sedikit kekurangan pada teknik lanjutan atau penerapan dalam skenario tertentu.	Menunjukkan pemahaman umum tentang proses dan strategi ETL, namun kurang mendalam pada metode optimisasi, penanganan kualitas data, atau skenario spesifik.	Menunjukkan pemahaman terbatas tentang strategi ETL, dengan beberapa miskonsepsi dan komponen penting yang hilang. *Demonstrates limited understanding of ETL strategies,*	Menunjukkan pemahaman yang sangat minim atau tidak ada tentang konsep atau proses ETL, dengan miskonsepsi besar atau penjelasan yang tidak relevan. *Shows little to no understanding of ETL concepts or*		

No	Related LOLOBJ-SO	Assessment criteria	Weight	Excellent (90 - 100)	Very Good (85-89)	Good (80-84)	Average (75-79)	Poor (0-74)	Score	(Score x Weight)
				penerapannya pada berbagai skenario. *Demonstrates thorough and in-depth understanding of ETL concepts, processes, and strategies, including advanced techniques, optimization methods, data quality handling, and tool selection, with the ability to explain their application in various scenarios.*	*Demonstrates clear and accurate understanding of ETL concepts, processes, and strategies, with minor gaps in advanced techniques or scenario application.*	*Demonstrates general understanding of ETL processes and strategies, but lacks depth in optimization methods, data quality handling, or specific scenarios.*	*with several misconceptions and missing key components.*	*processes, with major misconceptions or irrelevant explanations.*		
3	LO4	Kemampuan mendesain dan mengembangkan aplikasi Business	40%	Menunjukkan kemampuan yang sangat unggul dalam mendesain dan mengembangkan aplikasi BI yang lengkap, sangat fungsional,	Menunjukkan kemampuan yang kuat dalam mendesain dan mengembangkan aplikasi BI dengan model data terstruktur, KPI relevan, dan	Menunjukkan kemampuan memadai dalam mendesain dan mengembangkan aplikasi BI dengan dashboard fungsional dan KPI relevan,	Menunjukkan kemampuan terbatas dalam mendesain aplikasi BI, dengan model data yang tidak lengkap, analitik minimal, dan	Menunjukkan kemampuan sangat minim atau tidak ada dalam mendesain aplikasi BI, dengan kesalahan besar, fitur tidak lengkap, atau		

No	Related LOLOBJ-SO	Assessment criteria	Weight	Excellent (90 - 100)	Very Good (85-89)	Good (80-84)	Average (75-79)	Poor (0-74)	Score	(Score x Weight)
		Intelligence (BI)		mudah digunakan, dan efektif secara visual. Mengintegrasikan model data optimal, KPI yang akurat, analitik lanjutan, serta integrasi dengan sumber data relevan, memenuhi kebutuhan bisnis secara menyeluruh. *Demonstrates outstanding ability to design and develop BI applications that are complete, highly functional, user-friendly, and visually effective. Incorporates optimal data models, accurate KPIs, advanced analytics, and integration with relevant data*	dashboard fungsional, dengan sedikit kekurangan pada aspek kegunaan, kedalaman analitik, atau integrasi. *Demonstrates strong ability to design and develop BI applications with well-structured data models, relevant KPIs, and functional dashboards, with only minor shortcomings in usability, analytics depth, or integration.*	namun kurang dalam analitik, optimalisasi model, atau fitur lanjutan. *Demonstrates adequate ability to design and develop BI applications with functional dashboards and relevant KPIs, but lacks depth in analytics, optimal modeling, or advanced features.*	masalah kegunaan. *Demonstrates limited ability to design BI applications, with incomplete data models, minimal analytics, and usability issues.*	output tidak relevan. *Demonstrates little or no ability to design BI applications, with major errors, incomplete features, or irrelevant outputs.*		

No	Related LOLOBJ-SO	Assessment criteria	Weight	Excellent (90 - 100)	Very Good (85-89)	Good (80-84)	Average (75-79)	Poor (0-74)	Score	(Score x Weight)
				sources, meeting business requirements comprehensively.						
	Total Score: ∑(Score x Weight)									

Remarks:
Rubrik penilaian di atas berlaku secara utuh hanya jika kelompok mengumpulkan 4 jenis dokumen, yakni laporan, slide presentasi, dataset dan implementasi *project*-nya (dalam Power BI, Tableu atau Google Studio)

ASSESSMENT METHOD

Instructions :

- You are required to select a case study from a company (you may use data from your workplace or from public repositories).
- Choose and define 3 main business processes that you will address in your project.
- The case study and dataset you plan to work on must be approved by the course instructor no later than before the 3rd meeting.
- For the implementation of the data warehouse, ETL process, dashboard development, and BI, you are free to choose the tools (Power BI/Metabase, and others).
- Assignment 1, Assignment 2, and the Final Project must be completed individually.

Description:

1. Connecting visualization results with real business context.
2. Building a narrative (data storytelling) to convey insights from the data.
3. Providing strategic and actionable recommendations.

Deliverables:

1. Final Project Report (a compilation of reports from Assignment 1, Assignment 2, plus cover, Chapters 6–8, with an additional maximum of 20 pages, total maximum 70 pages) → structure as outlined below.
2. The dataset used.
3. ETL output files (e.g., .pbix with query, Pentaho .ktr, .py script, .sql query).
4. Data warehouse schema file (SQL DDL or diagram).
5. BI dashboard file.
6. Final Project presentation slides.
7. Final Project presentation video.

Sistematika Laporan:

1. **Cover**
2. **Kata pengantar**
3. **Daftar isi**
4. **Daftar gambar**
5. **Daftar table**
6. **Bab 1. Pendahuluan dan Profil Organisasi**
    1.1 Latar Belakang
    1.2 Profil Organisasi/Perusahaan
    1.3 Alasan Pemilihan Kasus
    1.4 Tujuan Pengembangan Data Warehouse
    1.5 Ruang Lingkup Proyek
7. **Bab 2. Perancangan Data Warehouse**
    2.1 Deskripsi Data
    - Sumber data (CSV, database, API, dll.)
    - Jenis data & karakteristiknya

- Permasalahan kualitas data

2.2 Entity Relationship Diagram (ERD)

- Diagram & penjelasan entitas, atribut, relasi

2.3 Analisis Proses Bisnis

- Minimal 3 proses bisnis
- Fakta & dimensi tiap proses
- Jenis proses (operasional/analitik/strategis)

2.4 Enterprise Bus Matrix

- Mapping fakta vs dimensi

2.5 Desain Star Schema

- Fact table, dimension table, hierarki

8. **Bab 3. Proses ETL**

3.1 Extract

- Identifikasi sumber data
- Metode ekstraksi (query/file/API)

3.2 Transform

- Minimal 3 langkah transformasi (cleaning, formatting, integration, aggregation)
- Tabel before vs after

3.3 Load

- Metode pemuatan data (full/incremental)
- Hasil load ke tabel fact/dimension

3.4 Tools yang Digunakan

- Power BI (Power Query) / Talend / Pentaho / SSIS / Python, dll.
- Screenshot workflow / script

9. **Bab 4. Desain Dashboard BI**

4.1 KPI Utama

- Definisi & alasan pemilihan KPI

4.2 Struktur Dashboard

- Halaman executive summary
- Halaman analisis detail

4.3 Jenis Visualisasi

- Pemilihan grafik/tabel
- Alasan sesuai data & KPI

10. **Bab 5 Interaktivitas Dashboard**

5.1 Filter, Drill-down, Slicer

5.2 Navigasi antar halaman

5.3 Contoh skenario penggunaan dashboard

11. **Bab 6. Data Story telling**

6.1 Pertanyaan Bisnis yang Dijawab

6.2 Narasi dari Data & Dashboard

6.3 Analisis Keterkaitan Antar Proses Bisnis

12. **Bab 7. Insight dan Rekomendasi Strategis**

7.1 Minimal 3 Insight Utama

7.2 Dampak terhadap bisnis (jawaban atas *so what?*)

7.3 Rekomendasi Action Plan

13. **Bab 8. Ringkasan Eksekutif**

- Ringkasan 1 halaman dengan bahasa non-teknis untuk stakeholder eksekutif

- Fokus pada insight & rekomendasi, bukan detail teknis

14. **Daftar Pustaka**
15. **Lampiran**
    a. Diagram dan skema
    b. Tangkapan layar dari proses ETL
    c. Workflow diagram ETL (misalnya di Pentaho/Talend/Power Query).
    d. Script Python/SQL (jika ada).

<u>Note for Lecturers:</u>

1. Assessment will be based on the completeness and quality of each component (report, slides, dataset, project implementation).
2. Full marks will be awarded if all components are submitted on time in accordance with the stated requirements.
3. If any component is incomplete or does not meet the requirements, the grade will be adjusted based on the applicable assessment rubric.

Assessment Activity Weight Score:

Assessment Activity	Weight
Assignment 1	30%
Assignment 2	30%
Final Project (AoL)	40%